Intelligent Candidate Discovery and Ranking

Senior AI Engineer shortlist for Redrob AI

Offline, explainable ranker optimized for production ML fit, behavioral availability, and trap resistance.

100K
candidate records streamed

Top 100
ranked shortlist output

3m 19s
full CPU ranking run

0 APIs
no network during ranking

Problem Framing

The challenge is not to find candidates with the most AI terms. The JD asks for a founding Senior AI Engineer who can own retrieval, ranking, evaluation, and product-facing ML systems.

What fails

- Keyword stuffing: long skill lists with weak career evidence.
- Inactive profiles: perfect on paper but not reachable.
- Off-domain AI: CV or speech profiles without retrieval/ranking depth.
- Consulting-only histories where the JD explicitly raises fit risk.

What should win

- Production retrieval, search, recommendation, and ranking systems.
- Evaluation rigor: NDCG, MRR, MAP, offline-online correlation, A/B tests.
- 5-9 year seniority band with hands-on coding evidence.
- Recent activity, strong response behavior, and practical logistics fit.

System Architecture

Input
Stream candidates.jsonl or .jsonl.gz

Feature Extraction
Parse profile, career history, skills, education, Redrob signals

Scoring
Combine JD evidence, behavior, logistics, and penalties

Selection
Maintain deterministic top-100 heap

Output
Write validated CSV plus audit JSON

Why this architecture

- Runs inside the 5 minute, CPU-only, no-network constraint.
- Uses interpretable scoring, so each candidate can be defended in manual review.
- Avoids per-candidate LLM calls and fragile black-box ranking behavior.

Ranking Method

Role/title fit	Rewards senior AI, search, recommendation, applied ML, NLP, senior data science roles.
Career evidence	Looks for production retrieval, ranking, vector search, hybrid search, RAG, evals, scale, latency.
Skill trust	Weights skills by proficiency, duration, endorsements, and Redrob assessment scores.
Behavior	Uses recency, response rate, response speed, open-to-work, interviews, notice period, GitHub, demand.
Logistics	Rewards India and target city fit; penalizes overseas candidates without relocation signal.
Trap penalties	Down-weights inconsistency, keyword stuffing, stale activity, non-technical roles, services-only fit.

Trap Resistance

The dataset includes traps. The ranker includes explicit guardrails for them.

Keyword stuffers

AI skills are trusted only when backed by duration, endorsements, assessment scores, or career evidence.

Honeypot-like profiles

Experience-year inconsistencies and duration mismatches are penalized before top-100 selection.

Inactive candidates

Last active date and recruiter response rate modify otherwise strong profile matches.

Wrong AI domain

CV/speech/research-only profiles need retrieval or ranking evidence to stay competitive.

Services-only careers

Service-company-only histories are down-weighted unless production evidence is unusually strong.

Non-technical titles

Business, HR, marketing, accounting, and support profiles cannot win on skills alone.

Behavioral Signals

Availability

- last_active_date
- open_to_work_flag
- notice_period_days

Responsiveness

- recruiter_response_rate
- avg_response_time_hours
- interview_completion_rate

Market signal

- profile_views_received_30d
- saved_by_recruiters_30d
- search_appearance_30d

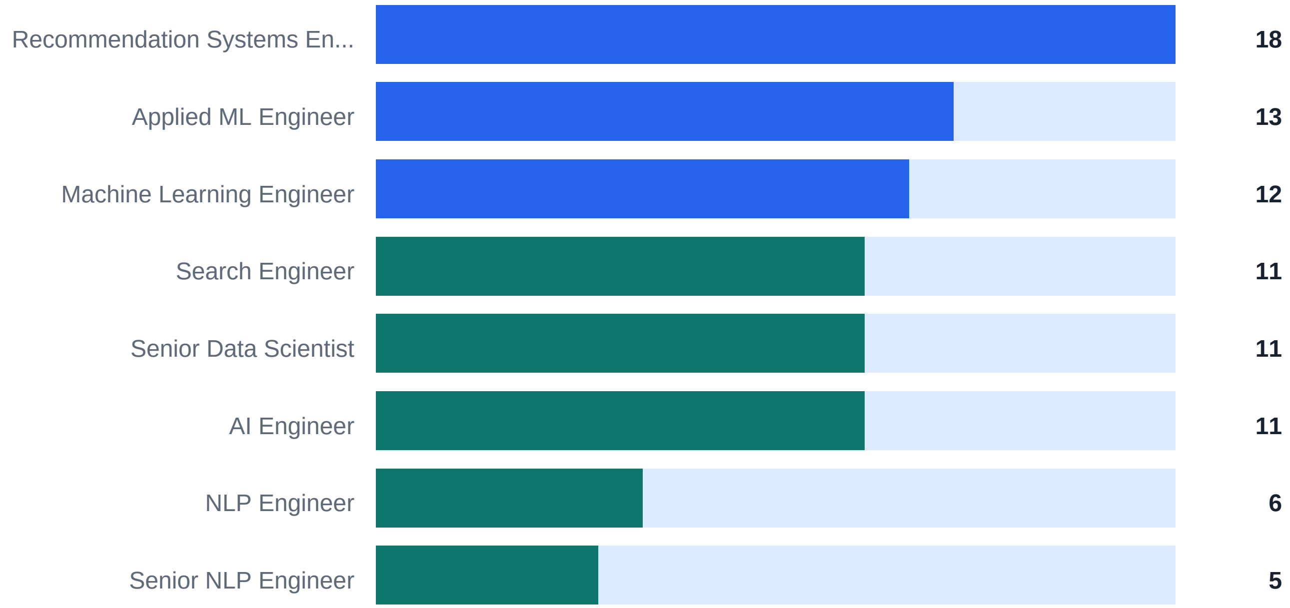
Trust signal

- verified_email
- verified_phone
- linkedin_connected
- github_activity_score

Behavior is used as a modifier: strong candidates with poor availability are pushed down, not automatically discarded.

Current Ranking Output

Top-100 title distribution



95

India-based candidates in top 100

0

services-only careers in top 100

0

stale profiles in top 100

Audit result: no non-technical keyword-stuffer titles and no flagged consistency traps in the generated top 100.

Top Ranked Candidates

Rank	Candidate	Title	Years	Location	Score
1	CAND_0077337	Staff Machine Learning Engineer	7.0	Kochi, Kerala	0.999527
2	CAND_0018499	Senior Machine Learning Engineer	7.2	Noida, Uttar Pradesh	0.999412
3	CAND_0081846	Lead AI Engineer	6.7	Jaipur, Rajasthan	0.999249
4	CAND_0046525	Senior Machine Learning Engineer	6.1	Pune, Maharashtra	0.999069
5	CAND_0064326	Search Engineer	7.6	Gurgaon, Haryana	0.999034
6	CAND_0002025	Senior AI Engineer	5.9	Trivandrum, Kerala	0.999010
7	CAND_0086022	Senior Applied Scientist	5.3	Kolkata, West Bengal	0.998943

Reasoning generation

Each row gets concise recruiter-facing reasoning grounded in candidate facts: title, years, location, evidence category, response rate, activity recency, notice period, and concerns.

Reproducibility

3m 19s
full ranking pass over 100,000 candidates

CPU only
no GPU inference or acceleration

Stdlib
no external runtime dependencies for ranking

Valid
passes validate_submission.py

```
python rank.py --candidates ./candidates.jsonl --out ./outputs/submission.csv
python validate_submission.py ./outputs/submission.csv
```

The ranking step streams input records, keeps a fixed top-100 heap, writes a CSV submission, and emits an audit JSON for review. It makes no hosted LLM or embedding API calls during ranking.

Why This Is Defensible

- It directly models the role: production retrieval, ranking, evaluation, and product ML systems.
- It uses behavioral signals because hireability is not just profile fit.
- It resists dataset traps by penalizing weak AI claims, stale profiles, non-technical keyword stuffers, and inconsistent histories.
- It is reproducible under the official constraints: CPU-only, no network, under 5 minutes.
- It produces auditable reasoning for manual review without hallucinating candidate facts.

Submission artifacts

outputs/submission.csv, src/ranker.py, README.md, docs/approach.md, output/pdf/redrob_ai_candidate_ranker_solution_deck.pdf